

3. Programming

- Programming is a transferable skill
- **Transferable skill:** skills developed in one situation which can be transferred to another situation.

3.2. Variables

- Declaring a variable:
 - Pseudocode: `"DECLARE ;"`
 - Python: no need to declare however must write above as a comment (`"python #..."`)

- Assigning variables:

`"python ← "` or `"python "`

`"python identifier = value"` or `"python expression"` or `"python "string""`

3.3. Selections

- "IF" Statement
 - Pseudocode: `IF...THEN...ELSE...ENDIF`
 - Python: `if (expression): (statements) else: (statements)`
- "CASE" Statement
 - Pseudocode: `CASE OF variable: ... OTHERWISE: ... ENDCASE`
 - Python: `if (expression): (statement) elif (expression): (statement) ... else: (statement)`

3.4. Iterations

Count-controlled Loop

Pseudocode	Python
FOR {identifier} <- {val1} TO {val2} STEP {val3}	{statement(s)}

Post-condition Loop

Pseudocode	Python
REPEAT	{statement(s)}

Pre-condition Loop

Pseudocode	Python
WHILE {condition}	{statement(s)}

3.5. Built-in Functions

String/character manipulation:

- Uppercase or lowercase all characters:
`("string").upper()` `("string").lower()`
- Finding length of a string: `len("string")`
- Converting:
 - String to Integer - `int("string")`
 - Integer to String - `str(integer)`

Random number generator: `random.randint(a, b)`

Where a and b defines the range

3.6. Benefits of Procedures and Functions:

- Lines of code can be re-used; don't have to be repeated
- Can be tested/improved independently of program
- Easy to share procedures/functions with other programs
- Create routines that can be called like built-in command

3.7. Procedure

Procedure: subroutine that performs a specific task without returning a value

- Procedure without parameters:

PROCEDURE <statement(s)>ENDPROCEDURE	def identifier():statement(s)
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- When a procedure has a parameter, the function can either pass it by either reference or value
- Pass by value:** data copied into procedure so variable not changed outside procedure

PROCEDURE <identifier> (BYVALUE <param>:
<datatype>)

<statement(s)>

ENDPROCEDURE

def identifier(param):

statement(s)

- Pass by reference:** link to variable provided so variable changed after going through procedure (*not in Python*)

PROCEDURE <identifier> (BYREF <param>:
<datatype>)

<statement(s)>

ENDPROCEDURE

- Calling a procedure:

CALL ()	Identifier()
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3.8. Function

Function: subroutine that performs a specific task and returns a value

Functions are best used to avoid having repeating blocks of code in a program, as well as increasing the reusability of code in a large program.

FUNCTION <identifier> (<parameter>: <data type>)
RETURNS <datatype>

<statement(s)>

ENDFUNCTION

def identifier(param):

statement(s)

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return expression
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